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TRANSPORT AND RETENTION OF BACTERIOPHAGES IN TWO TYPES OF WILLOW-CROPPED LYSIMETERS

Key Words: Bacteriophage, willow coppice, lysimeters, groundwater pollution

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ABSTRACT

Irrigation and fertilization of short-rotation willow coppice with wastewater is a new way of reusing wastewater in Sweden. To evaluate the possible impact of viruses on groundwater quality, the transport and retention of the bacteriophage *Salmonella Typhimurium* type 28B were studied in two types of willow-cropped field lysimeters containing clay or sand soil. Phages were applied to the soil surface and moderate irrigation was done daily under field-like conditions.

In the clay, soil rapid transport of bacteriophages was recorded with breakthrough at 1.2-m depth after 2–24 hours indicating macropore flow through the soil. Phage transport through the sand soil varied considerably, but was in general much slower and the phage retention much higher compared with the clay soil. The willow plants were not found to facilitate phage leaching. Instead, the results indicate the presence of phage retaining processes in the rhizosphere.

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INTRODUCTION

The reuse of wastewater for irrigation of forest and agricultural crops has been practised in many countries for decades. In Sweden, there is a growing interest for wastewater irrigation of energy crops, mainly short-rotation willow coppice (SRWC). This alternative wastewater treatment method is adopted to improve nitrogen treatment efficiency at a reasonably low cost (Rosenqvist et al., 1997), and at the same time reuse water and plant nutrients for increased biomass productivity.

The sanitary aspects of land application of wastewater are crucial when comparing this method with conventional techniques for wastewater treatment. Especially pathogenic viruses in wastewater may constitute a problem due to the potentially high numbers present and the low dose needed for infection (Oron et al., 1995). Therefore, elucidating the transport behaviour of viruses in soils is of major importance for evaluating the potential sanitary risks of wastewater irrigation systems. The transport of viruses in the soil-water matrix can be rapid, but varies depending on, for example, soil characteristics, types of virus and climate (Keswick and Gerba, 1980; Yates et al., 1985). Viruses reaching the groundwater will have an increased survival time due to the low temperature in the groundwater (Yates et al., 1985).

Bacteriophages as tracers of the transport and retention of human viruses in soil has been used in several studies (Bales et al., 1989; Jin et al., 1997; McKay et al., 1993a). The advantage of using bacteriophages is that they mimic the behaviour of potentially pathogenic viruses and have a relatively good survival rate. They allow dilution ratios up to 10^{-12} due to the large numbers that can be propagated, are harmless to humans and to the environment, and can easily be detected and quantified (Havelaar et al., 1991).

In our study the bacteriophage *Salmonella Typhimurium* type 28B (Lilleengen, 1948) was used. This phage has been used as a model virus in several tracer studies (Johansson et al., 1998; Stenström, 1995). When using bacteriophages for

tracing polluted sources is it important to choose a phage that naturally will not be found in the wastewater or sludge. No phages have been detected in analysed control samples. The aim of our study was to determine virus leaching patterns and retention in an irrigated short-rotation willow coppice system under field-like conditions using a bacteriophage as a model for animal viruses potentially present in domestic wastewater.

Two types of gravity-drained lysimeters were used comprising eight large (willow-cropped) lysimeters with disturbed clay and sand profiles, and six small lysimeters (four willow-cropped and two non-cropped) with undisturbed sand profiles. In the small lysimeters we were able to study the effect of the willow plant on the phage leaching pattern. Phage applications were made on two occasions during two consecutive years.

MATERIAL AND METHODS

Lysimeters

In 1988, cylindrical fibre glass lysimeters were permanently installed at a lysimeter station on the Ultuna campus in Uppsala, Sweden (59°49'N, 17°39'E). The mean annual temperature and precipitation in the area were 5.5°C and 527 mm, respectively. Each lysimeter was provided with a plastic pipe at the base for collection of leachate water (Figure 1), which was distributed to a below-ground station. In 1989, the lysimeters were filled with two types of soil. When filling the lysimeters a layer of gravel was first placed in a funnel at the bottom of each lysimeter container. Sub- and topsoil from two nearby arable fields were then used to re-establish soil profiles in the lysimeters. This was done by carefully packing about 20 cm of soil at a time to obtain a bulk density which closely resembled the conditions at the sites where the soils were collected. Four lysimeters (1c-4c) were filled with Kungsängen clay soil (post-glacial alluvial clay) and four (5s-8s) with Nântuna sand (postglacial alluvial sand). The Nântuna sand has a 0.25 m deep topsoil layer with a clay content of 8% enabling some aggregate formation, while

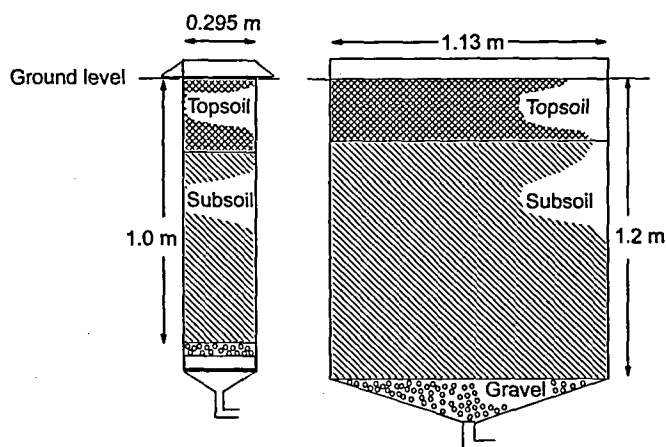


FIGURE 1

Design of the Small and Large Lysimeters Installed at the Lysimeter Station in Uppsala, Sweden.

the subsoil is structureless. The Kungsängen clay has a well-developed structure with stable aggregates and permanent cracks down to at least 1.0 m depth and a topsoil depth of 0.23 m. The properties of the two undisturbed soils were described by Wiklert et al. (1983) (Table 1). The soil depth, surface area and volume of each lysimeter were 1.2 m, 1.0 m^2 and 1200 l, respectively. After installation, weeds established in the lysimeters and no farm practices were carried out until 1997. In May 1997, weeds in the lysimeters were treated with glyphosate and 10 days later the topsoil was tilled manually and all plant tissue was removed.

Six undisturbed soil columns of the Nântuna sand were collected in May 1997 using a tractor-mounted hydraulic soil auger. Cylindrical PVC pipes with inner diameters of 0.295 m and lengths of 1.18 m were used as lysimeter containers (Figure 1). During collection, a container was inserted in a cylindrical steel pipe with a cutting head which rotated around the casing, pushing a soil core into the casing as it moved down through the soil. Undisturbed soil columns were taken down to 1.10-m depth. The technique is described in detail by Persson and

TABLE 1
Soil Properties of the Sand and the Clay Soils (from Wiklert et al., 1983)

	Näntuna sand		Kungsängen clay	
	Topsoil	Subsoil	Topsoil	Subsoil
Soil type (USDA)	Sandy loam	Loamy sand	Silty clay	Silty clay
Clay content (%)	8	1	45	49
Organic matter content (%)	2	1	10	6
Dry bulk density (g/cm ³)	1.47	1.46	1.05	0.90
Porosity (%)	44	46	58	65

Bergström (1991). After collecting the soil columns, 0.10 m of soil was removed at the base of each lysimeter and replaced with a nylon mesh followed by a gravel layer, a porous plastic sheet and a fibreglass lid with holes allowing gravity drainage. The depth, surface area and volume of the undisturbed soil columns were 1.0 m, 0.068 m² and 68 l, respectively. The lysimeters were placed in pipes permanently installed in the ground at the lysimeter station. Silicone adhesive was used to prevent air fluxes in the small gap between the casing wall and the installation pipe. A fibreglass cap was also placed on the upper edge of the casing to protect the small gap from rainfall. When installed in their pipes, the lysimeter casings stood on bowl-shaped collectors connected to the belowground station via plastic pipes, and the soil surface in the lysimeters corresponded with the ground level of the surroundings. Before installation all living plant tissue was removed from the topsoil. The thickness of the topsoil was about 0.30 m in all lysimeters.

In June 1997, 20-cm cuttings of *Salix viminalis* (clone 78183) were planted in the lysimeters; two cuttings 0.5 m apart in each of the eight large lysimeters, and one cutting in each of four (A-D) of the six small lysimeters. Two of the small lysimeters (E & F) functioned as controls during the 1998-year phage application and were not planted. All lysimeters were kept weed free by manual weeding. An automatic irrigation system distributed water and plant nutrients to the lysimeters. Water pumps were connected to two 1000-l plastic tanks in which a balanced plant nutrient solution (BLOMSTRA, Cederroth International AB; N:P:K-ratio 51:10:43) was mixed with tap water to reach fixed nitrogen concentrations (18

and 37 mgN/l, respectively). The fertilizer solution was distributed to the lysimeters via drip irrigation pipes. The daily supply of the solution to each large lysimeter was 6.0 mm at a flow rate of 16 mm/h via four drippers, and each small lysimeter received 44 mm of solution at a flow rate of 117 mm/h via two drippers. Irrigation and fertilization started on 30 June 1997.

Phage Application

The bacteriophage *Salmonella Typhimurium* type 28B (Lilleengen, 1948) with a size of 60 nm according to electron microscopic measurements (K-O Hedlund, pers. comm. SIIDC, 1998) was used in the experiment. The bacteriophage *Salmonella Typhimurium* type 28B is very stable, in low temperatures (5°C) the phage can survive for more than 10 years. The bacteriophage was propagated in nutrient broth (NB, Oxoid) against its host strain *Salmonella Typhimurium* type 5 to a concentration of around 10^{10} plaque forming units (pfu)/ml. The recipient strain was killed by adding chloroform (10 ml/l). The solution was then centrifuged for 30 min at 4300 x g and finally filtered through 0.45- μ m membrane filters.

The first phage application was made on 8 October 1997. At that time drainage occurred at a flow rate of approximately 7 mm/day and 40 mm/day from the large and small lysimeters, respectively; i.e. the water saturation level was above field capacity. A phage solution with a concentration of 1.0×10^{10} pfu/ml was mixed with 150 l tap water in a 1000-l water tank to give an initial phage concentration of 3.4×10^7 pfu/ml. Using the drip irrigation system, a 1.4 mm pulse was applied to each of the four large lysimeters (two with clay soil; 1c & 2c; and two with sand soil; 5s & 6s) and a 9.5 mm pulse to two willow-cropped small lysimeters (A & B). The phage dose applied to the large and small lysimeters was 5.2×10^{10} and 1.1×10^{10} pfu, respectively. After applying the phage pulse, the irrigation system was washed and no significant extra input of phages followed the initial pulse. After the phage application, 6.0 mm of tap water was applied to the large lysimeters daily for six days (total of 36 mm). For the small lysimeters, 44 mm water with fertilizers was applied daily until 30 November (total of 2332 mm). In

spring 1998, irrigation and fertilization of the small lysimeters was resumed on 30 March and lasted until 29 June (total of 3960 mm). Irrigation and fertilization of the large lysimeters was resumed on 22 April and lasted until mid August (total of 660 mm).

The second phage application was made on 21 April 1998. Again, drainage occurred from the lysimeters at a flow rate of approximately 0.4 mm/day and 40 mm/day from the large and small lysimeters, respectively; i.e. the water saturation level was above field capacity. The procedure was similar to that of the preceding year. A phage solution with a concentration of 4.0×10^{10} pfu/ml was mixed with 250 l tap water in one 1000-l water tank to give a final phage concentration of 2.4×10^8 pfu/ml. A 6.0 mm pulse was then applied as before to four new large lysimeters (two with clay soil; 3c & 4c; and two with sand soil; 7s & 8s) and a 44 mm pulse to four new small lysimeters (two willow-cropped; C & D; and two non-cropped; E & F) using the drip irrigation system. The total dose applied to the large and small lysimeters was 7.8×10^{11} and 3.9×10^{10} pfu, respectively. After the phage pulse, the irrigation system was washed with tap water and no significant extra input of phages followed. The large lysimeters were thereafter irrigated daily with 6.0 mm of water with fertilizer until mid August (total of 660 mm) and the small lysimeters with 44 mm until 29 June (total of 3080 mm).

Sampling and Analyses

The drainage water from each lysimeter was collected in the below-ground lysimeter station. Samples were collected from some of the the lysimeters using automatic samplers with 24 one-litre bottles. Each bottle was weighed for determination of drainage volume and a 60 ml subsample was taken in a polythene bottle with a screw cap. Before re-installation in the sampler the bottles were washed in cold tap water, boiling water and with a disinfectant 1% VIRKON (Antec Int. Ltd) solution, and finally with tap water. Some carry-over of phages, although small (estimated to be less than 0.1%), was likely to occur in the automatic sampler due to its construction. Drainage from the other lysimeters was collected in 23-litre polythene barrels. The barrels were weighed for determination

of drainage volume and water samples were taken in 60-ml polythene bottles with screw caps. The barrels were washed with tap water before being re-installed and the carry-over was estimated to be less than 0.01%. All samples were stored for 1-6 weeks in darkness in the below-ground lysimeter station at +5 to +8 °C before being transported to the Swedish Institute for Infectious Disease Control for analysis. Phage deactivation during storage was found to be negligible (1 log₁₀ reduction after 30 weeks).

Three plaque-assays with different detection limits were used, depending on the concentration of the phages in the sample. For all methods, the bacteriophages were analysed against the host strain *Salmonella Typhimurium* type 5, cultured in nutrient broth at 37°C for 4 hours (Adams, 1959). For high concentrations, a spot-test was used; Petri dishes with CM 55 agar (Bloodagar base, Oxoid) were soaked with the host strain and dried, and for each dilution of the sample four 10-µl samples were spotted on the plate. For lower concentrations a double-agar-layer (DA) method were used; one millilitre of the sample was mixed with 0.5 ml of the host strain and 8.5 ml of a soft agar (60% nutrient broth (NB)+ 40% nutrient agar (NA)). Two millilitres of the mixture were spread on a dry CM 55 (NA) agar plate. For low concentrations of phages a 10-ml water sample was analysed after mixing with 1 ml of the host strain and 10-15 ml of CM 55 agar (NA) and then poured on two or three Petri dishes. The plates were incubated at 37°C for 18 hours and the clear zones were then counted and calculated as plaque forming units (pfu). The minimum detectable level for the 10-millilitre method was 0.1 pfu/ml.

RESULTS

Large Clay Lysimeters

The transport of phages through the lysimeters was rapid after both applications. After the 1997 application, phages were recorded after 6-12 hours in the intensively sampled lysimeter (1c) and within 24 hours in the daily sampled lysimeter (2c) (Figure 2; Table 2). Peak concentrations were recorded after 125

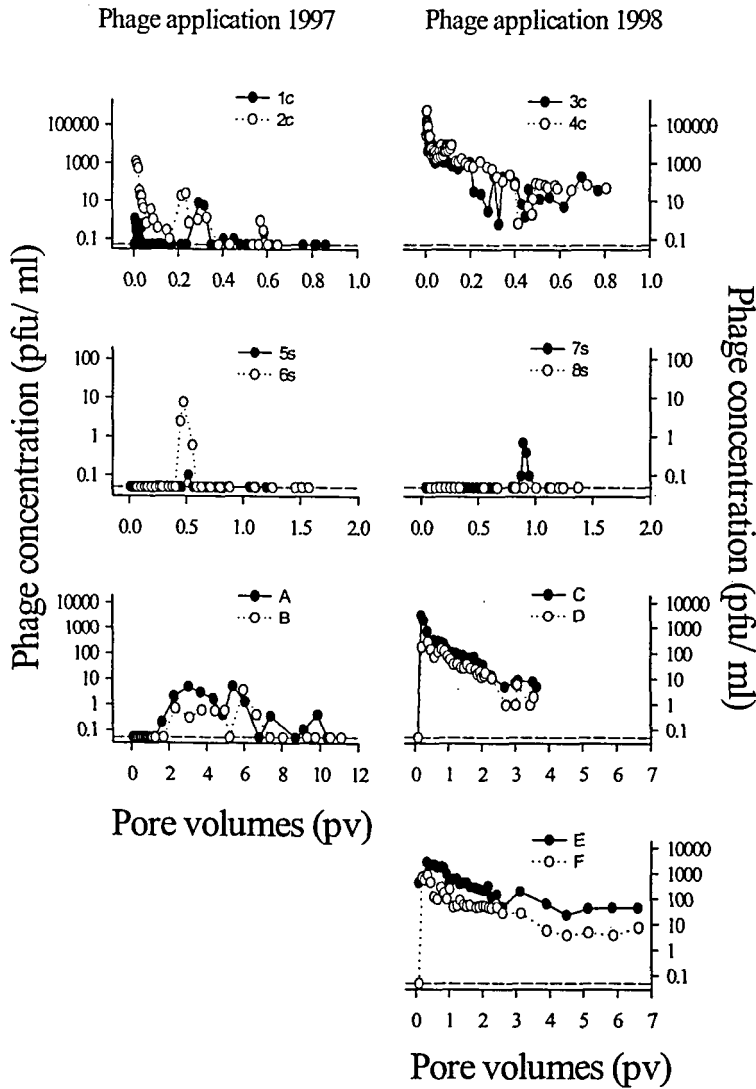


FIGURE 2

Phage Concentrations in Drainage Water from Large Clay (1c-4c) and Sand (5s-8s) and Small Sand (A-F) Lysimeters after Phage Application in October 1997 and April 1998, Respectively. Reference Line Represents Non-detectable Levels.

TABLE 2
Summary of Results from Phage Experiments 1997 and 1998

Lysimeter No.	Date of phage application	No. of phages applied (pfu)	Break through time and (discharged pore volumes)	Time at peak conc.	Peak conc. (pfu/ml)	Accum. phage transport (pfu)	Accum. phage transport, (%)
<i>Large clay</i>							
1c	8 Oct-97	5.2×10^{10}	<12 hr (3.79×10^{-3})	125 d	7.7	4.36×10^5	8.4×10^{-4}
2c	8 Oct-97	5.2×10^{10}	<1 d (7.58×10^{-3})	1 d	1.2×10^3	1.41×10^7	2.7×10^{-2}
3c	21 Apr-98	7.8×10^{11}	<1 d (7.32×10^{-3})	3 d	9.8×10^3	2.79×10^8	3.6×10^{-2}
4c	21 Apr-98	7.8×10^{11}	<2 hr (7.98×10^{-4})	4-6 hrs	5.8×10^5	2.22×10^9	2.8×10^{-1}
<i>Large sand</i>							
5s	8 Oct-97	5.2×10^{10}	159 d (0.52)	159 d	0.1	1.91×10^3	3.7×10^{-6}
6s	8 Oct-97	5.2×10^{10}	126 d (0.45)	128 d	7.7	2.10×10^5	4.0×10^{-4}
7s	21 Apr-98	7.8×10^{11}	246 d (0.88)	251 d	0.7	6.41×10^2	8.2×10^{-8}
8s	21 Apr-98	7.8×10^{11}	nd.	-	-	0	0
<i>Small sand</i>							
A	8 Oct-97	1.1×10^{10}	21 d (1.65)	180 d	5.1	3.80×10^5	3.5×10^{-3}
B	8 Oct-97	1.1×10^{10}	27 d (2.35)	180 d	3.6	1.51×10^5	1.4×10^{-3}
C	21 Apr-98	3.9×10^{10}	2 d (0.21)	2 d	3.2×10^3	2.62×10^7	6.7×10^{-2}
D	21 Apr-98	3.9×10^{10}	2 d (0.21)	4 d	2.8×10^2	4.75×10^6	1.2×10^{-2}
E	21 Apr-98	3.9×10^{10}	<1 d (0.11)	4 d	2.9×10^3	7.14×10^7	1.8×10^{-1}
F	21 Apr-98	3.9×10^{10}	2 d (0.21)	4 d	9.0×10^2	1.44×10^7	3.7×10^{-2}

days (1c) and 1 day (2c), corresponding to a discharge of 0.3 and 7.6×10^{-3} pore volumes (pv), in the respective lysimeter. The cumulative number of phages transported through the lysimeters differed substantially (Table 2). Phages could be detected in the drainage water until 223 (1c) and 434 days (2c) after phage application, respectively.

After the phage application in 1998, breakthrough of phages was recorded within 2 hours in the intensively sampled lysimeter (4c) and within 24 hours in the daily sampled one (3c), corresponding to a discharge of less than 8.0×10^{-4} and 7.3×10^{-3} pv, respectively (Figure 2; Table 2). Peak concentrations were higher than during the preceding application and were reached after 4-6 hours (3.4×10^{-3} pv) and 3 days (2.6×10^{-2} pv), respectively. The total phage transport through the lysimeters was also higher and corresponded to 0.28 and 0.034% of what was

applied, respectively. Phages could still be detected in the drainage from both lysimeters after 352 days when the last sampling was done.

Large Sand Lysimeters

In the large sand lysimeters the phage transport was much slower and the total amounts of phages transported through the lysimeters were much less than in the clay lysimeters. After the application in 1997, phages in drainage water could only be detected at very low concentrations on one (5s) and three (6s) occasions, respectively, and these were in connection with snow melting (Figure 2; Table 2). The total phage transport through the lysimeters was $3.7 \cdot 10^{-6}$ and $4.0 \cdot 10^{-4}$ % of what was applied, respectively. Phages could be detected until 159 days after phage application in lysimeter 6s.

After the application in 1998, the pattern was similar. Phages were detected on four occasions at low concentrations in one (7s) and not at all in the other lysimeter (8s) (Figure 2; Table 2). The proportion of total phage transport through lysimeter 7s was $8.2 \cdot 10^{-8}$ %, and phages could be detected in drainage water until 259 days after application.

Small Undisturbed Sand Lysimeters

After application in October 1997, the phage retention of the soil-plant system was substantial (Figure 2; Table 2). Phage breakthrough was recorded after 21 (A) and 27 (B) days, respectively, corresponding to a discharge of 1.65 and 2.35 pore volumes. The peak phage concentrations, which were recorded 180 days after application, were low, 5.1 (A) and 3.6 (B) pfu/ml. The amounts of water which had leached through the lysimeters by this time corresponded to 5.4 and 6.0 pv, respectively. Phages could be detected in drainage water until 250 (A) and 187 (B) days after application, respectively. The cumulative number of phages transported through the lysimeters was low and corresponded to $3.5 \cdot 10^{-3}$ and $1.4 \cdot 10^{-3}$ % of the total phage input to lysimeter A and B, respectively.

After the phage application in April 1998, the leaching pattern differed markedly from the previous year (Figure 2; Table 2). Phage breakthrough was already recorded after 2 days in the willow cropped lysimeters (C and D),

corresponding to a discharge of 0.2 pv. The peak phage concentrations were also much higher than after the 1997 phage application, and they were reached within 2-4 days. The proportions of total phage transport through the lysimeters were approximately one order of magnitude higher than after the application in 1997. There were no clear differences between the willow-cropped (C and D) and the non-cropped lysimeters (E and F). Phages could be detected in drainage until the lysimeters were destructively sampled 69 days after phage application.

DISCUSSION

The results clearly demonstrate the variation in phage leaching patterns in the different soils. The very rapid phage transport in the large clay lysimeters, with breakthrough within hours after surface application, could only be explained by substantial macropore flow of water through the soil. This is in accordance with what has been found in other studies of virus transport in well-structured soils (McKay et al., 1993a; McKay et al., 1993b; Sinton et al., 1997; McMurphy et al., 1998). McMurphy et al. (1998) found a significant correlation between the transport pattern of water and faecal coliforms (bacteria) in a soil prone to preferential flow. The macropores responsible for most of the water transport also accounted for most of the transport of the coliforms. The disturbance of the soil when filling the large lysimeters probably caused increased macropore formation, but the long time-span between lysimeter installation and our study (8 years) should to a large extent have normalized the soil properties. Therefore, the properties of the soil in the lysimeters, for example, as regards presence of macropores, were probably similar to those of the undisturbed soil, and the phage leaching pattern found in the lysimeters could be valid for a field situation.

Rapid transport of viruses could lead to groundwater contamination. When reaching the groundwater zone, the virus survival time can be prolonged due to low temperatures and absence of UV-light (Keswick and Gerba, 1980; Yates et al., 1985). The leaching pattern after phage application in 1998 was similar to that after phage application in 1997, but phage concentrations in the discharge were

roughly one order of magnitude higher. This could be due to the higher dose and concentration of phages applied in 1998, which, however contradicts the findings of Lance et al. (1982; see below).

The phage retention in the large sand lysimeters was very high after both applications. Indeed, the accumulated number of phages leached was several orders of magnitude lower than the corresponding number of phages leached from the clay lysimeters. This contradicts the results from several other studies, where clayey soils have been found to be more efficient in retaining viruses than sandy and organic soils (Keswick and Gerba, 1980, Sinton et al., 1997). The phage breakthrough after 0.45-0.88 pore volumes indicates a convective/dispersive transport in the sand soil. The leaching pattern agrees well with that found by Jin et al. (1997) for the phages MS-2 and ϕ X-174.

The accumulated number of phages leached from the large sand lysimeters after application in 1998 was no higher than after the application in 1997, despite the one order of magnitude larger dose applied. This indicates not only that the phage retention was higher, but also that phage leaching was not substantially affected by the larger dose and higher concentration of phages applied 1998, which is in accordance with findings by Lance et al. (1982). Therefore, the very different leaching pattern in the small undisturbed sand lysimeters after the two applications was probably not explained by the different amount of phages applied in 1998 compared with 1997. Indeed, the marked difference in leaching pattern is difficult to explain. Initially after phage application, daily discharge and climatic conditions did not differ markedly, but the plants were in different growth phases. After the spring application of phages (April 1998), breakthrough of phages was recorded within two days. This indicates preferential flow of water through the soil. Since there were no clear differences between the willow-cropped and the non-cropped (control) lysimeters in this respect, the willow plants did not seem to facilitate leaching of phages in the sand soil, e.g. by creating root channels allowing preferential flow of water. One possible explanation for the very clear differences between phage retention after autumn and spring

application would be differences in the rhizosphere (root-soil boundary) properties affecting phage sorption. Plant roots are known to leach hydrocarbons, exudates, which affect the root tip environment and supply microbial biomass with carbon. During winter the carbon supply from the roots diminishes, which could affect the rhizosphere composition, and possibly also the size and composition of the microbial biomass. How this in turn would affect the phage retention is unclear. Unfortunately, as there were no control lysimeters during the phage application in 1997, no conclusions could be drawn about the extent of phage sorption in the rhizosphere of actively growing plants.

The differences between the small and large sand lysimeters in spring 1998 are also difficult to explain. The irrigation rate was higher for the small lysimeters (44mm/d) as compared with the large lysimeters (6mm/d), which should have increased the degree of water saturation and also affected the flow pattern of water in the lysimeters (Bergström and Shirmohammadi, 1999). In several studies, the retention of virus in soils have been found to be higher during non-saturated flow conditions compared with saturated flow conditions (Keswick and Gerba, 1980; Powelson et al., 1990; Powelson and Gerba, 1994), possibly due to virus inactivation at the air-water interface (Poletika et al., 1995). This might be a possible explanation, but the very marked differences in phage leaching patterns were most probably due to some other (unknown) factor.

CONCLUSIONS

The results show that the retention of virus can differ dramatically between different soil types and that macropore flow in a clay soil can facilitate very rapid transport of virus to groundwater. This calls for precaution when locating and designing systems for wastewater irrigation of crops. Despite the rapid transport of viruses through leaching, overall retention was fairly high in the clay soil. In the sand soil, virus retention, although varying, could be extremely high, but preferential flow of water through sand soil can facilitate rapid virus breakthrough. The willow plants were not found to facilitate virus transport

through the soil. In one case, virus retention was unexpectedly high, which may have been due to unknown processes in the rhizosphere. This should be subject to further studies.

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REFERENCES

- Adams, M.H. "Bacteriophages". Interscience Publishers Ltd., London (1959).
- Bales, R.C., Gerba, C.P., Grondin, G.H., and Jensen, S.L., Applied and environmental microbiology, 55(8), 2061-2067 (1989).
- Bergström, L.F. and Shirmohammadi, A., J. Soil Contamination, 8(6), 637-651 (1999).
- Havelaar, A.H., Butler, M., Farrah, S.R., Jofre, J., Marques, E., Ketranakul, A., Martins, M.T., Ohgaki, S., Sobsey, M.D., Zaiss, U., Wat. Res. 25, 529-545 (1991).
- Jin, Y., Yates, M.V., Thompson S.S. and Jury, W.A., Environ. Sci. Tech., 31, 548-555 (1997).
- Johansson, P-O., Espeby, B., Nilsson, B. and Allestam, G., In Artificial Recharge of Groundwater. (Eds by Peters, J.H. et al.) Balkema AA. Rotterdam. pp 383-385. (1998).
- Keswick, B.H. and Gerba, C.P., Environ. Sci. Tech., 14, 1290-1297 (1980).
- Lance, J.C., Gerba C.P. and Wang, D-S., J. Environ. Qual. Vol., 11 (3), 347-351 (1982).
- Lilleengen, K., "Typing of Salmonella Typhimurium by means of bacteriophage." Dissertation, The Bacteriological Hygienical Department of the Royal Veterinary College, Stockholm (1948).

McKay, L.D., Cherry, J.A., Bales, R.C., Yahya, M.T., & Gerba, C.P., Environ. Sci. Tech., 27, 1075-1079 (1993 a).

McKay, L.D., Gillham, R.W. and Cherry, J.A., Water Resources Research., 29 (12), 3879-3890 (1993 b)

McMurry, S.W., Coyne, M.S. and Perfect, E., J.Environ. Qual., 27, 86-92 (1998)

Oron, G., Goemans, M., Manor, Y. and Feyen, J., Wat. Res., 29 (4), 1069-1078 (1995).

Persson, L., and Bergström, L., Soil Sci. Soc. Am. J. 55, 285-287, (1991).

Poletika, N.N., Jury, W.A. and Yates M. V., Water Resources Research., 31 (4), 801-810 (1995).

Powelson, D.K., Simson, J.R. and Gerba, C.P., J.Environ. Qual., 19, 396-401 (1990).

Powelson, D.K. and Gerba, C.P., Wat. Res., 28 (10), 2175-2181 (1994).

Rosenqvist, H., Aronsson, P., Hasselgren, K. and Perttu, K., Biomass and Bioenergy., 12(1), 1-8 (1997).

Sinton, L.W., Finlay, R.K., Pang, L. and Scott, D.M., Water, Air and Soil Pollution. 98, 17-42 (1997)

Stenström, T.A., In: Conference Proceedings Safe Water Environments, Kenya, 21-25 August, 1995. Report 24, p 73-78. Linköping University, Sweden, Department of Water and Environmental Studies (1995).

Wiklert, P., Andersson, S. and Weidow, B., "Studier av markprofiler i svenska åkerjordar." En faktsammanställning (in Swed.). Report 132, Swedish University of Agricultural Sciences, Department of Soil science, Division of Agricultural Hydrotechnics. (1983).

Yates, M.V., Gerba C.P. and Kelley. L.E., Applied and Environmental Microbiology. 49, 778-781 (1985).